

Predicting Diabetes from Lifestyle and Demographic Using Support Vector Machines

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Research Question

- Can self reported behavioral and demographic variables from a national household survey classify adults ever diagnosed with diabetes?
- Does adding non-linear kernels (radial or polynomial) improve on a linear boundary- and which variables matter the most?

Introduction

Diabetes is one of the most prevalent chronic disease in the United States, affecting roughly 1 in 10 adults and carrying significant health and economic consequences. Understanding which lifestyle and demographic factors predict diabetes diagnosis is clinically valuable for risk assessment to guide the public.

This analysis used data from 2022 National Health Interview Survey (NHIS), this was accessed through IPUMS Health Survey, to predict whether adult has ever been diagnosed with diabetes based on 16 self-reported variables spanning from demographic, physical activity, diet, sleep, and social status. Support vector machine (SVM) were suited to this task- they helped find an optimal separating hyperplane in high-dimensional feature space, through kernel trick, model non-linear decision boundaries without explicitly constructing new features.

This research used radial, linear, and polynomial to assess whether non-linear boundaries improve classification and identify which variables carry the most predictive weight. The linear kernel also provided interpretable feature weights, offering insight into the direction and relative magnitude of each predictor's contribution to the decision boundary.

Variables

- Demographic: age, sex(binary), partnered, education(ordinal 0-4), income group(ordinal 0-2)
- Body/Health: BMI(continuous), sleep hours per night
- Physical Activity: moderate activity min/day, vigorous activity min/day
- Diet: Fruit, vegetable, soda, fries, pizza, alcohol, work hours.

Methodology

Data were drawn from 2022 NHIS with target group of individuals from 18-85. The cleaned data provided sample with 14,164 adults with complete records across all 16 predictors. Diabetes prevalence in the cleaned sample was 12.8%

The outcome was DIABETICEV, a binary indicator of whether the respondent has ever been told by doctor that they have diabetes. Ambiguous responses were excluded, variables was recorded as Yes (2) or No(1).

To address the class imbalance, the majority class was downsampled to 1:1 ratio, yielding a balanced dataset of 3,614 observations. Data were split 75/25 into training (n=2712) and test (n=902) sets using stratified sampling. All continuous predictors were standardized to mean zero using parameters from training dataset.

Three SVM kernels were fit by using e1071 in R studio. The linear kernel found a flat hyperplane in the original feature space. Radial kernel maps features to higher dimension space by using Gaussian function (tune on C and bandwidth γ). Polynomial kernel captured feature interactions up to degree d (tune on C, d , and bias term r). All hyperparameters were selected via 5-fold cross validation.

Models were evaluated by using accuracy, sensitivity specificity, F1 score, and AUC. Subgroups were analyzed as well, with focusing in SEX (Male vs Female) and Older adults(50+ individuals). Feature weight was performed for linear kernel to assess each predictor's direction and relative importance.

Results

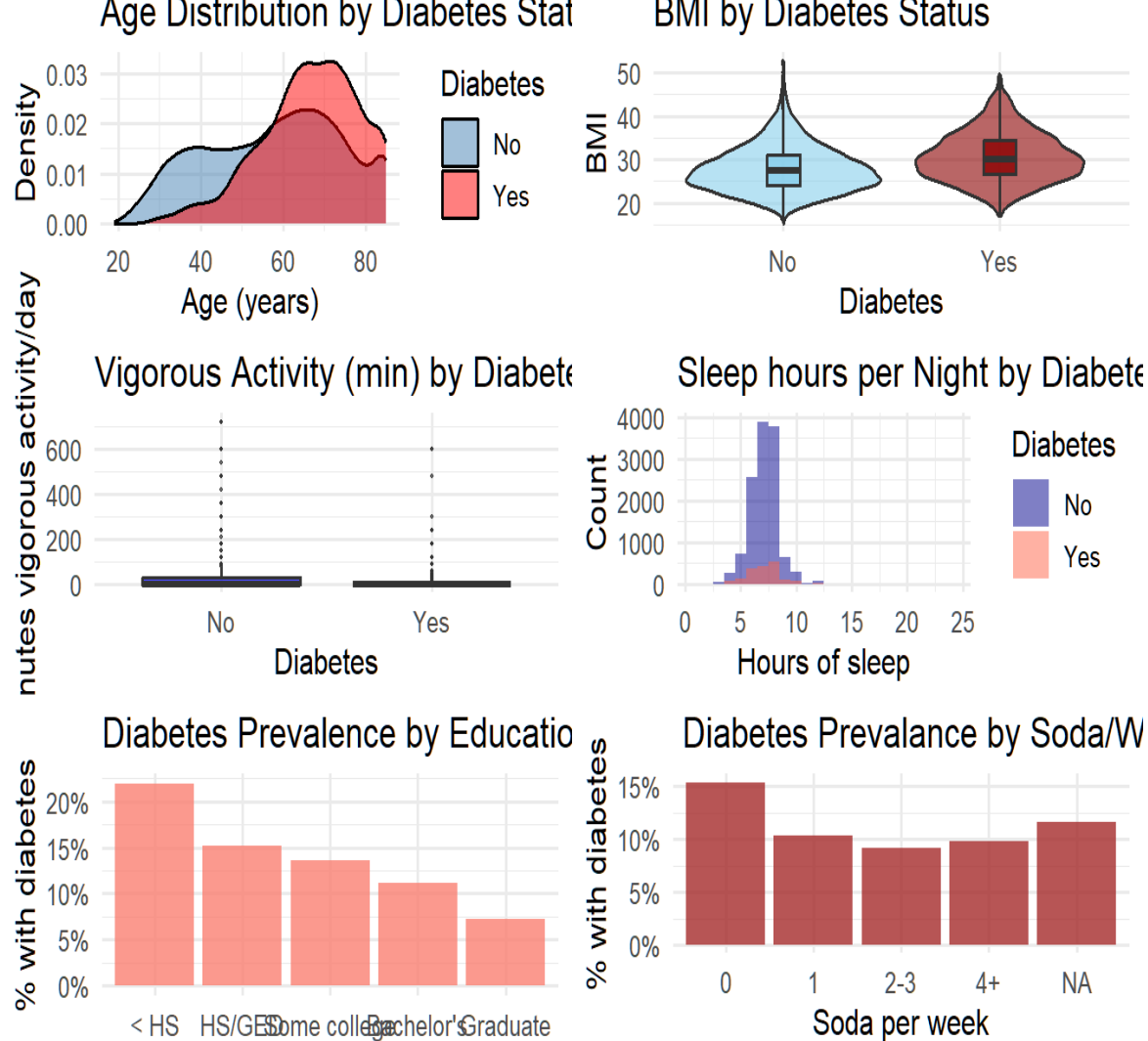


Figure 1: Health and Lifestyle Patterns by Diabetes status

All three SVM kernels achieved similar accuracy, the Polynomial kernel consistently provides better recall and F1 performance. This is matter more when positive class is the priority of the dataset.

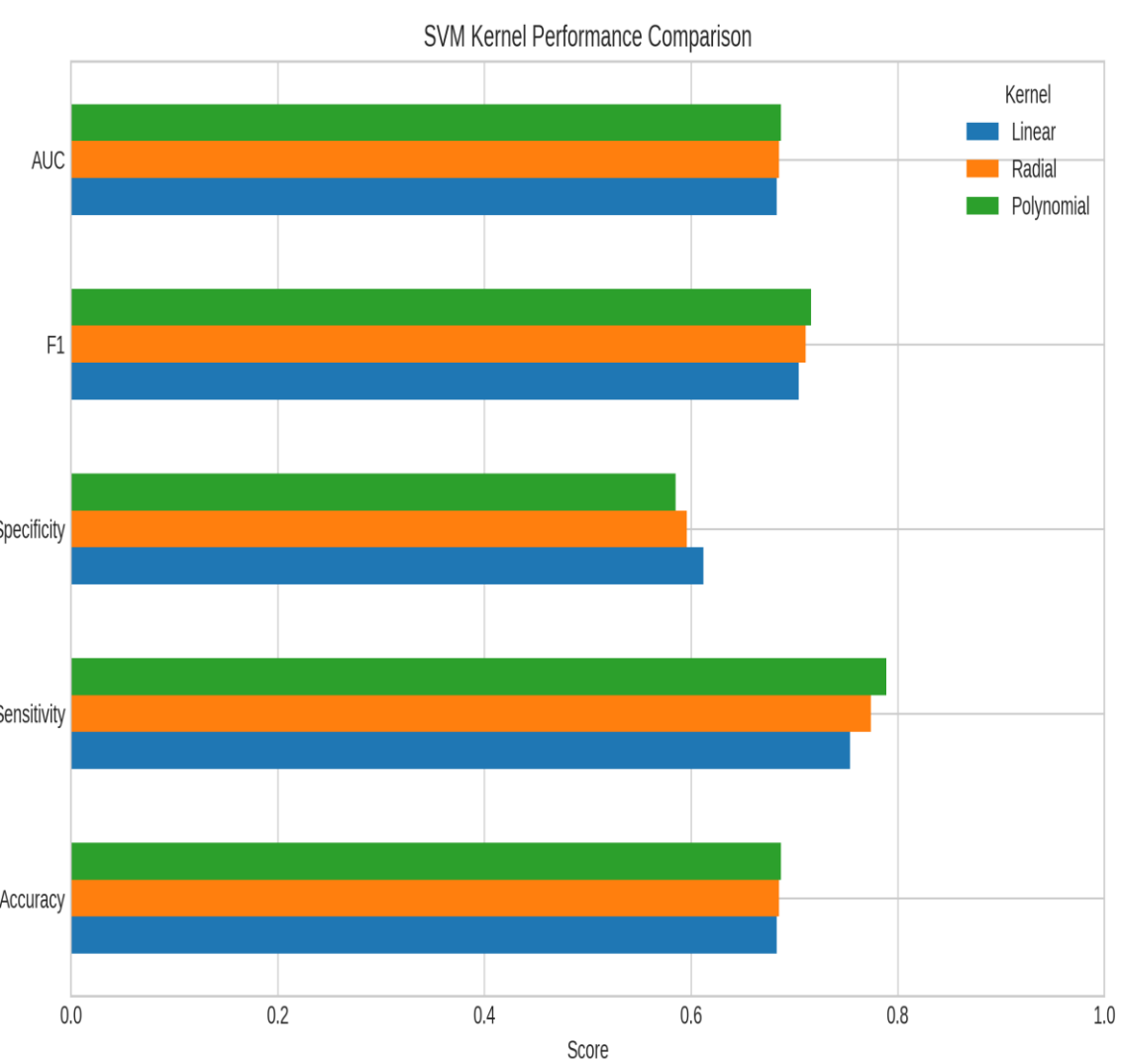


Figure 2: SVM Kernel Performance Comparison

All three kernels produced nearly identical ROC curve (AUC range 0.683-0.687), this indicated that the decision boundary for this problem is largely linear in the scaled feature space. The polynomial kernel (degree=2, C=0.1) edged ahead by modeling pairwise interaction, such as Age vs BMI. With AUC approximately 0.69, it is difficult to predict a complex disease trend from self-report survey data itself.

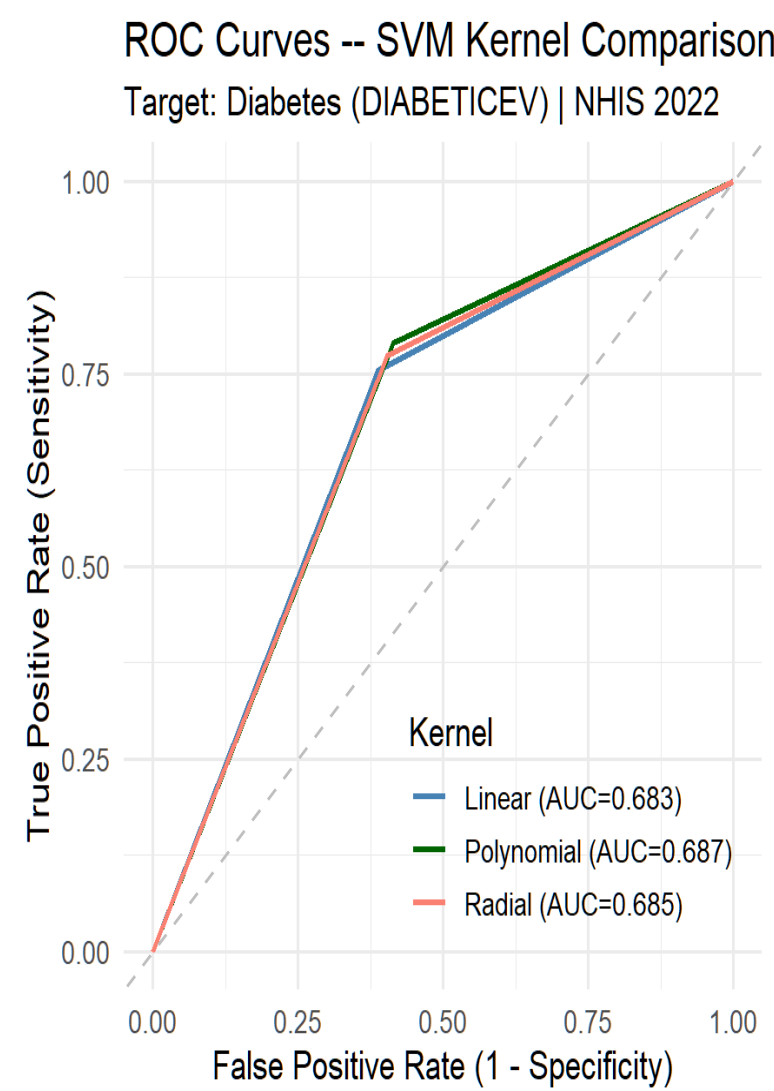


Figure 3: ROC Curves- SVM Kerel Comparison

Conclusion

In conclusion, this analysis demonstrated that self-reported survey from NHIS can classify diabetes diagnosis with moderate accuracy using support vector machines. The polynomial kernel performed the best across all metrics (68.7% accuracy, F1 score of 0.716, sensitivity 78.9%). Though, the narrow margin from this kernel only slightly over linear and radial, this indicated the decision boundary is largely linear in the standardized feature space.

Age and BMI were the strongest predictors, aligning with know diabetes risk factors. Vigorous activity showed a clear protective pattern, while education and income displayed strong inverse gradients in diabetes prevalence. Sleep contributed little discriminative value.

The modest AUC (~0.69) reflects the difficult of predicting a multifactorial disease from self-reported survey data and the limitations of cross-sectional design, recall bias, and missing clinical variables.

Overall, behavioral and socioeconomic factors provide meaningful, but limited predictive signal, and SVMs are reasonable modeling choice.

Recommendations

- Probability calibration should be explored.
- Survey weights should be incorporated.
- Test for alternative disease outcomes.

Acknowledgements

- Data: NHIS 2022 via IPUMS Health Survey
- Template: Posternerd